CINEMETRICS

Statistical Analysis of Financial & Critical Success Drivers in Global Cinema

Project Report

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1. INTRODUCTION

The global film industry represents a complex intersection of creative expression and commercial strategy, where artistic vision, financial investment, and audience reception collectively determine a movie's success or failure. With production budgets often exceeding hundreds of millions of dollars, stakeholders increasingly rely on data-driven insights to minimize financial risk and maximize both critical and commercial impact.

In recent years, the availability of large-scale movie datasets and advancements in data science have enabled deeper analysis of cinematic performance. However, most existing studies remain limited to descriptive analytics, focusing only on ratings or box-office collections in isolation. There is a clear need for an integrated, end-to-end analytical framework that combines financial, technical, and creative factors while also extending into predictive modeling.

This project, CINE-METRICS: The Anatomy of a Blockbuster, addresses this gap by systematically analyzing real-world movie data and developing machine learning models to predict cinematic success across multiple business-driven categories.

1.1 Project Background

Traditional movie success metrics like box-office and critic reviews are often analyzed in isolation, failing to capture the multidimensional nature of the industry. This project utilizes structured metadata from **IMDb and TMDb** to analyze over 1,500 films through a quantitative lens. By executing the full **Data Science Lifecycle**—from data engineering to supervised machine learning—the study moves beyond simple observation to provide a predictive framework for cinematic success.

1.2 Problem Statement

The film industry faces high uncertainty regarding profitability and reception. Existing analyses often lack statistical rigor or treat success as a simple binary (hit/flop). This project addresses these gaps by answering:

- Does high budget correlate with high returns or ratings?
- How do genre and talent impact financial outcomes?
- Can success be predicted using only pre-release metadata?

We resolve these by defining nuanced **success categories**, applying **hypothesis testing**, and developing **predictive models**.

1.3 Motivation and Real-World Relevance

As studios and streaming platforms shift toward **data-driven decision-making**, there is a critical need for analytics to guide budgeting, marketing, and talent selection.

Real-World Impact:

- **Risk Assessment:** Evaluating high-budget investments.
- **Strategic Planning:** Optimizing genre selection and release timing.
- **Predictive Insights:** Forecasting performance prior to release.

Academic Contribution: This project demonstrates an end-to-end integration of statistical inference and machine learning using complex, multi-source datasets to interpret artistic success through a business-centric, analytical framework.

2. PROJECT OBJECTIVE

2.1 Overall Aim of the Study

The primary aim of this study is to analyze and model real-world movie data to understand the key financial, technical, and creative factors that influence cinematic success. The project seeks to bridge artistic quality and commercial performance using data-driven methods.

2.2 Core Analytical Goals

- Examine the impact of budget, revenue, genre, runtime, director reputation, and cast strength on:
    - Return on Investment (ROI)
    - Audience ratings
- Identify success patterns and archetypes using exploratory data analysis, statistical inference, and dimensionality reduction techniques.

2.3 Predictive Modeling Objectives

- Develop supervised machine learning models to classify movies into:
    - A detailed 6-class success taxonomy
    - A simplified 3-class business-oriented success taxonomy
- Evaluate whether a movie's success category can be predicted using structured metadata alone.

2.4 Data Science Lifecycle Followed

The project strictly follows the Data Science Lifecycle:
Data Collection → Data Cleaning → Exploratory Data Analysis → Statistical Inference → Feature Engineering → Dimensionality Reduction → Predictive Modeling → Interpretation
.

3. DATASET STRATEGY AND DATA INTEGRATION

3.1 Data Collection Philosophy

This project is grounded in the use of real, publicly verifiable data, avoiding any form of synthetic or simulated datasets. The objective is to ensure analytical validity and real-world relevance by integrating multiple authoritative data sources. The dataset is designed to capture both the artistic and commercial dimensions of cinema, enabling reliable statistical analysis and predictive modeling.

3.2 Primary Data Source: IMDb Dataset

The primary dataset is derived from the IMDb Non-Commercial Dataset, consisting of over 1,500 movies. This dataset provides core movie attributes such as:

- Movie identifiers and titles
- Genres and spoken language
- Runtime and release information
- User ratings and vote counts

IMDb serves as the foundational source for audience engagement metrics and technical metadata, forming the backbone of the analytical workflow.

3.3 Secondary Data Enrichment using TMDb API

To incorporate the business and personnel dimensions missing from IMDb, the dataset is enriched using the TMDb API. This enrichment adds:

- Financial data: budget and revenue
- Creative attributes: director and top cast details
- Production-related information

The IMDb dataset is programmatically merged with TMDb data using unique movie identifiers, resulting in a unified and enriched dataset suitable for advanced analysis and modeling.

3.4 Final Dataset Schema and Feature Description

The final integrated dataset consists of 15+ features, categorized as follows:
Identification

- movie_id
- original_title

Categorical Features

- genre
- spoken_language
- popularity
- director
- top_casts
- production_companies

Numerical Features

- budget_usd
- revenue_usd
- runtime_minutes
- vote_average (user_rating)
- vote_count
- release_date

Derived Targets

- roi_percentage
- success_category (6-class)
- success_3class (3-class)

4. SUCCESS LABELING STRATEGY

4.1 Six-Class Success Taxonomy

To capture the full spectrum of cinematic performance, movies are classified into six industry-relevant categories:

All-Time Blockbuster, Blockbuster, Super Hit, Hit / Semi-Hit, Average / Below Average, and Flop / Disaster.

This taxonomy enables fine-grained analysis of financial outcomes and audience reception.

4.2 Business Interpretation of Success Categories

The six-class structure reflects real-world box-office performance levels, distinguishing between moderate success, high profitability, and commercial failure. It allows meaningful comparison across genres, budgets, and release strategies while preserving business realism.

4.3 Three-Class Target Engineering

For predictive modeling stability, the original categories are consolidated into a simplified three-class target:

- **Flop**: Flop / Disaster, Average / Below Average
- **Hit**: Hit / Semi-Hit, Super Hit
- **Blockbuster**: Blockbuster, All-Time Blockbuster

This transformation improves class balance and model interpretability.

4.4 Rationale for Target Consolidation

The three-class formulation reduces noise, enhances generalization, and aligns with practical decision-making scenarios, enabling reliable prediction of movie success using structured pre-release metadata.

5. DATA PREPROCESSING AND CLEANING

5.1 Data Merging and Integration

The IMDb dataset was merged with TMDb API data using unique movie identifiers. This integration unified ratings, financial details, and personnel information into a single analytical dataset.

Some Unnecessary columns were dropped like:

'status', 'backdrop_path', 'homepage', 'overview', 'poster_path', 'tagline', 'production_countries', 'spoken_languages'

Check for null values and found no null values then splitted the date into year and month column for separate visualization according to year and months or season wise analysis.

A new calculated column (roi_percentage) was included in the dataset which is measured by this formula:

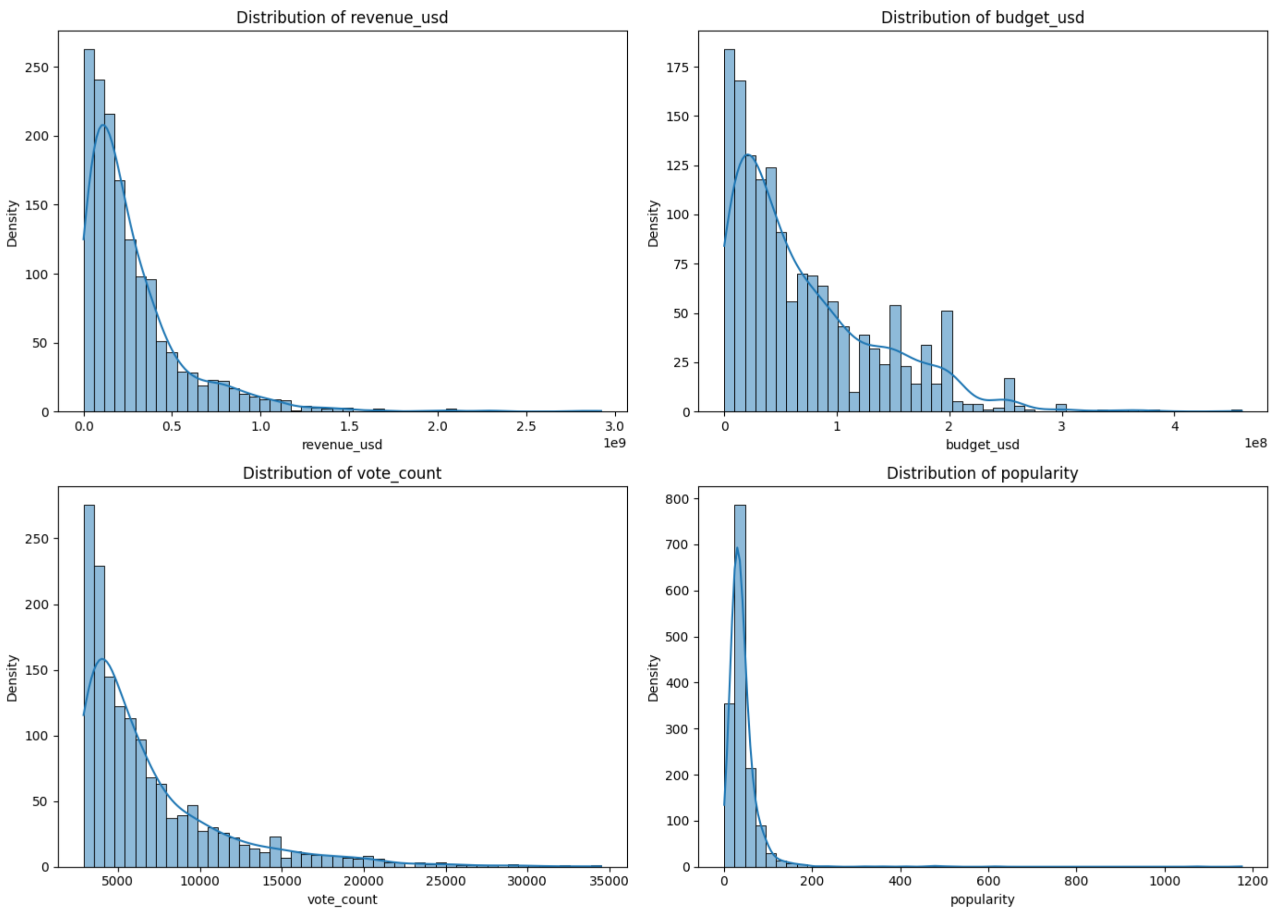
ROI (%) = ((Revenue_USD − Budget_USD) / Budget_USD) × 100

One more derived column from roi_percentage is the success_category which is categorized on the basis of roi_percentage:

Verdict	Standard ROI Range (Profit / Cost)	What it Means
Flop / Disaster	ROI <0%	The distributor/producer lost money on the investment.
Average / Below Average	ROI 0% to 50%	The investment recovered its cost (Break-Even), possibly with a small profit, or just under the break-even point.
Hit / Semi-Hit	ROI 50% to 100%	The investment earned back its cost plus a reasonable profit, often covering the high risk involved.
Super Hit	ROI 100% to 200%	The film doubled its investment, providing excellent returns.
Blockbuster	ROI 200% to 400%	The film delivered exceptionally high returns, making 3 to 5 times the investment.
All-Time Blockbuster	ROI >400%	A phenomenal success that redefines records.

5.2 Data Cleaning, Outlier Handling and Data Validation

Extreme values in budget and revenue were examined using the **IQR method**.



The dataset contains extreme values that reflect the heavy-tailed nature of the movie industry rather than data errors. Therefore, outliers were retained and handled using robust techniques such as log transformation and percentile capping (for ROI) instead of removal.

Count of outliers after applying log transformation:

revenue_usd : 109

budget_usd : 34

vote_count : 106

popularity : 113

roi_percentage : 143

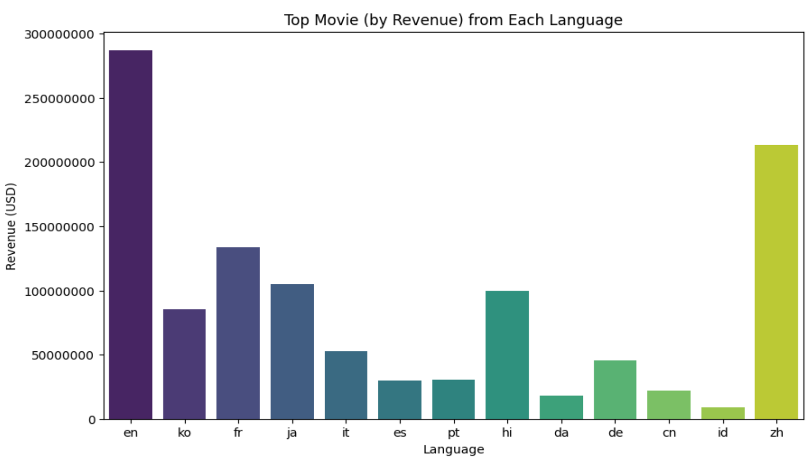
Despite applying log transformation and percentile capping, a small number of observations are still flagged by the IQR method. These represent legitimate extreme cases inherent to the movie industry and were intentionally retained to preserve real-world variability.

5.4 Final Clean Dataset Overview

After preprocessing, the final dataset consisted of **1,500+ cleaned and enriched movies**, ready for exploratory analysis, statistical inference, dimensionality reduction, and predictive modeling.

6. EXPLORATORY DATA ANALYSIS (EDA)

6.1 Top Movie (by Revenue) from Each Language

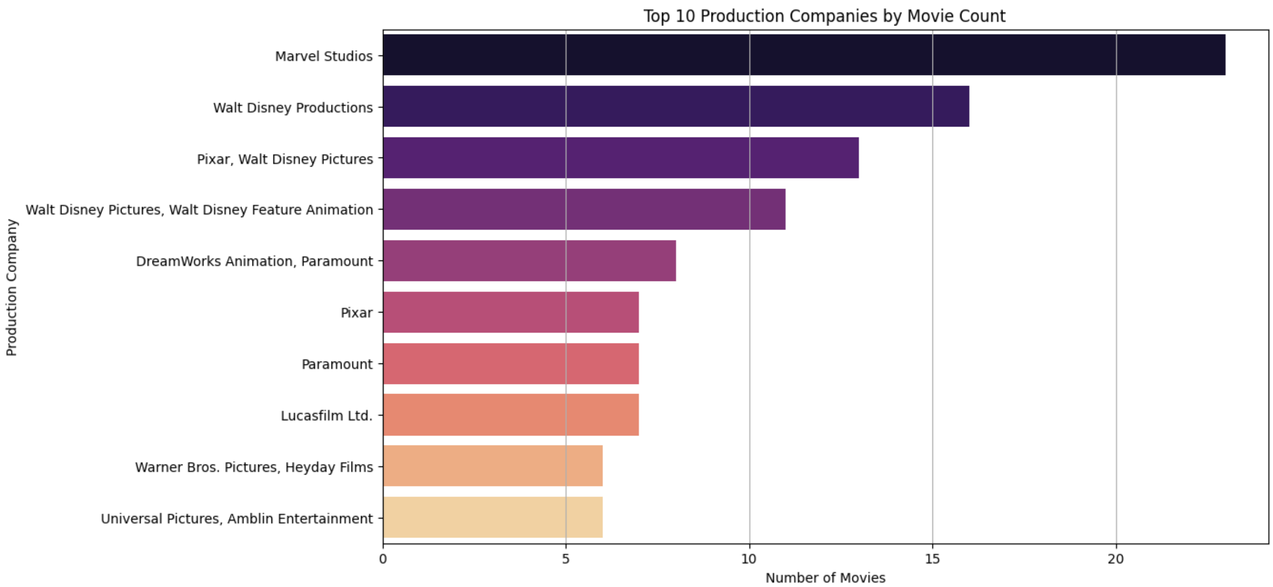



```python
plt.figure(figsize=(10, 6))
sns.barplot(
    x='original_language',
    y='revenue_usd',
    data=df,
    palette='viridis',
    hue='original_language',
    legend=False,
    errorbar=None
)
```

Insights

- English-language films generate the highest revenues globally.
- Chinese and Hindi films also show strong blockbuster-level earnings.
- Revenue is highly concentrated in a few top-performing movies per language.
- Language plays a significant role in commercial reach and profitability.

6.2 Top 10 Production Companies by Movie Count



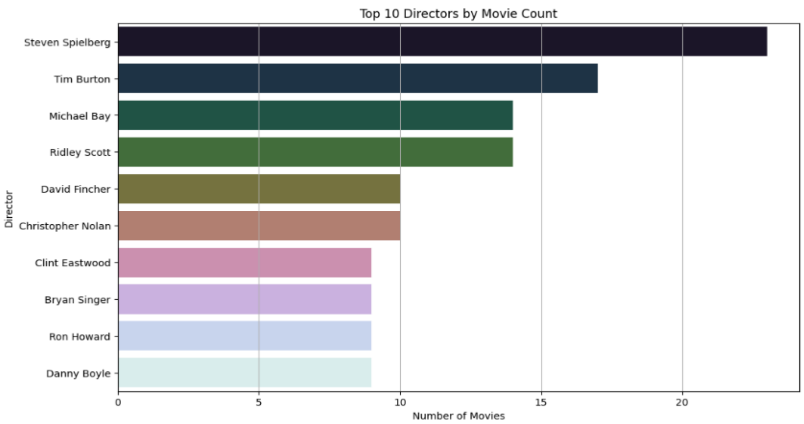
Insights

- Marvel Studios and Walt Disney dominate in production volume.
- Major studios consistently appear, indicating market consolidation.
- High movie count reflects strong production capacity and franchise-driven output.
- Studio scale is a key factor in sustained commercial presence.

Key Code Snippet

```
plt.figure(figsize=(12, 7))

top_production_companies = df['production_companies'].value_counts().head(10)

sns.barplot(
    x=top_production_companies.values,
    y=top_production_companies.index,
    hue=top_production_companies.index,
    palette='magma',
    legend=False
)
```

6.3 Top 10 Directors by Number of Movies

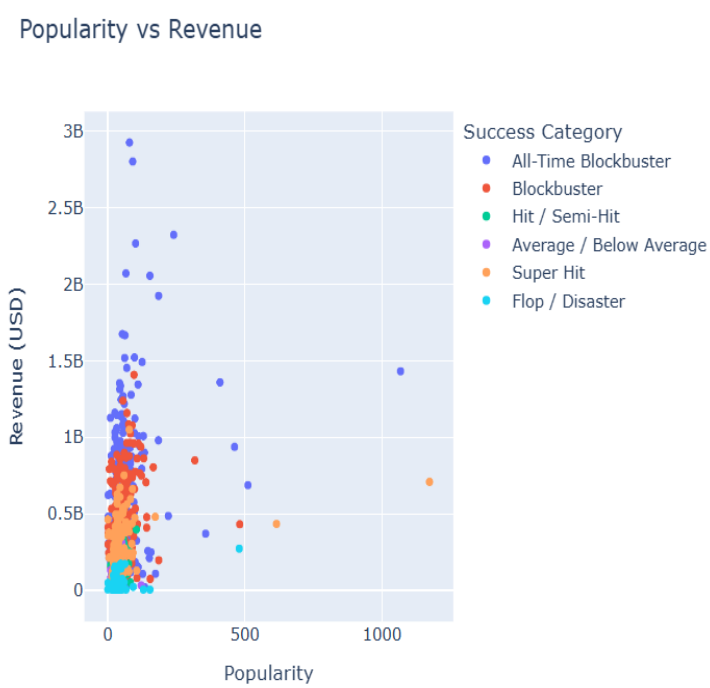



```python
plt.figure(figsize=(12, 7))

top_directors = df['director'].value_counts().head(10)

sns.barplot(
    x=top_directors.values,
    y=top_directors.index,
    hue=top_directors.index,
    palette='cubehelix',
    legend=False
)
```

Insights

- Steven Spielberg leads in movie count, reflecting sustained industry influence.
- A small group of directors consistently produce high numbers of films.
- High output indicates strong studio trust and commercial reliability.
- Director experience emerges as a meaningful success-related feature.

6.4 Popularity vs Revenue

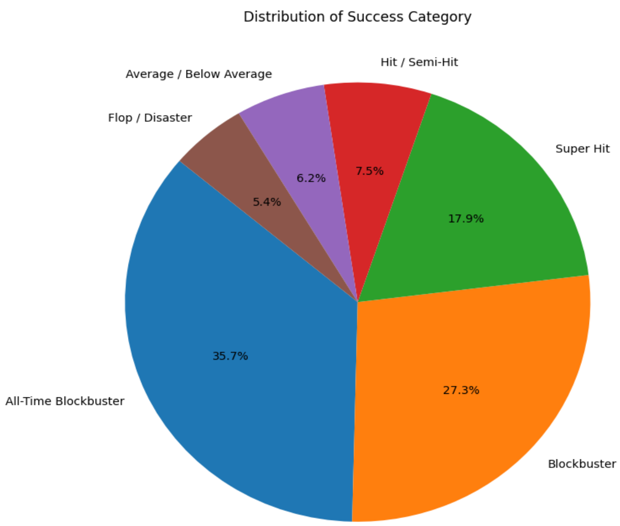



```python
import plotly.express as px

fig = px.scatter(
    df,
    x='popularity',
    y='revenue_usd',
    color='success_category',
    title='Popularity vs Revenue',
    labels={
        'popularity': 'Popularity',
        'revenue_usd': 'Revenue (USD)'
    }
)

fig.update_layout(
    legend_title='Success Category',
    xaxis_title='Popularity',
    yaxis_title='Revenue (USD)'
)

fig.show()
```

Insights

- Higher popularity generally aligns with higher revenue, but the relationship is non-linear.
- Blockbusters cluster at high revenue even with moderate popularity values.
- Flop categories remain concentrated at low revenue regardless of popularity.
- Popularity alone is insufficient to predict financial success.

6.5 Distribution of Success Category



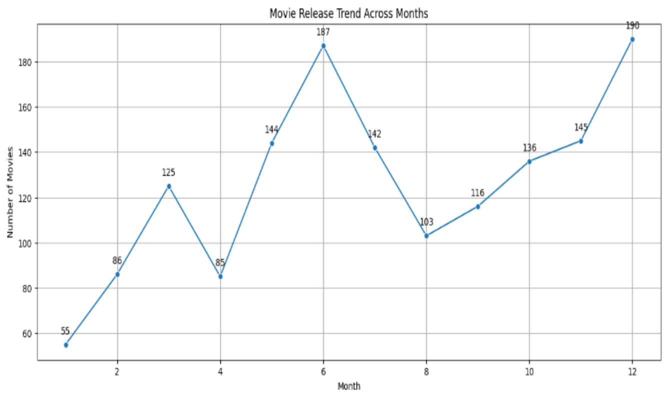

```python
# Count success categories
success_counts = df['success_category'].value_counts()

plt.figure(figsize=(8, 8))

plt.pie(
    success_counts.values,
    labels=success_counts.index,
    autopct='%1.1f%%',
    startangle=140
)
```

Insights

- A large proportion of movies fall under **Blockbuster** and **All-Time Blockbuster**, indicating revenue concentration among top performers.
- **Super Hit** movies form a significant middle tier, bridging commercial success and audience acceptance.
- **Flop / Disaster** and **Average / Below Average** categories together represent a smaller share, highlighting survivorship bias in the dataset.
- The imbalance across categories justifies the creation of a **3-class success label** for stable predictive modeling.

6.6 Temporal Trends

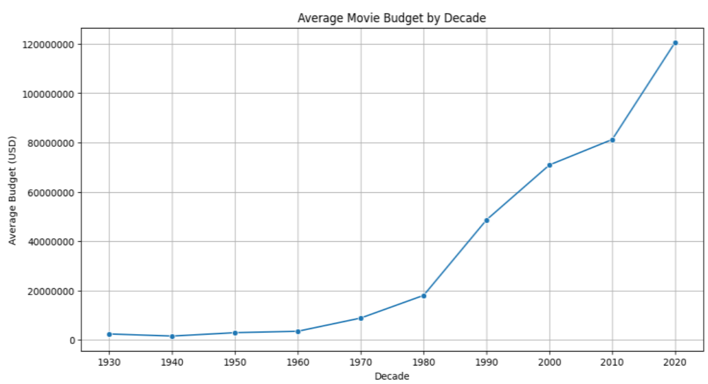
A. Movie Release Trend Across Months



Insights

- Movie releases peak during mid-year (May–July) and year-end (December).
- These periods align with summer blockbusters and holiday releases.
- Fewer releases occur in early months, indicating strategic scheduling.

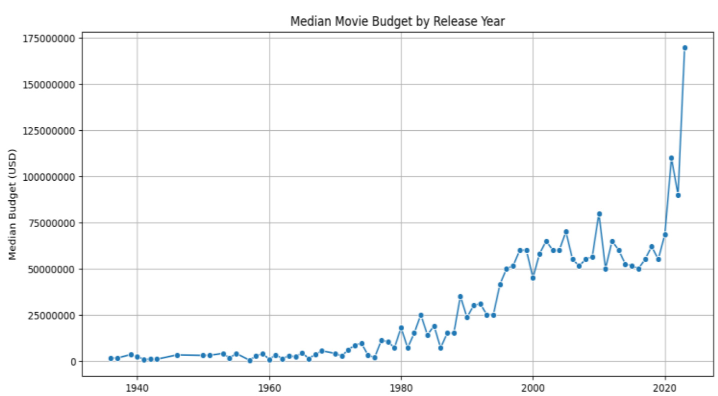
B. Average Movie Budget by Decade



Insights

- Average movie budgets show a strong upward trend across decades.
- Post-1980s, budgets increase sharply, reflecting higher production and marketing costs.
- Modern cinema is increasingly capital-intensive.

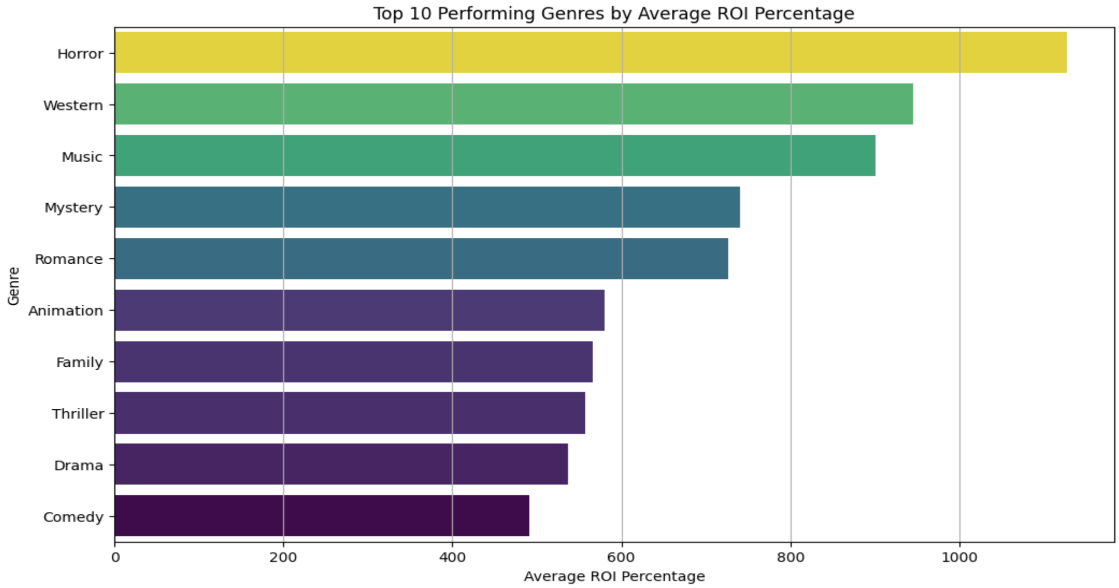
C. Median Movie Budget by Release Year



Insights

- Median budgets rise steadily over time, confirming industry-wide budget inflation.
- Budget growth accelerates after the 1990s, coinciding with franchise-driven cinema.
- The gap between low- and high-budget films widens in recent years.

6.7 Top 10 Genres by Average ROI Percentage



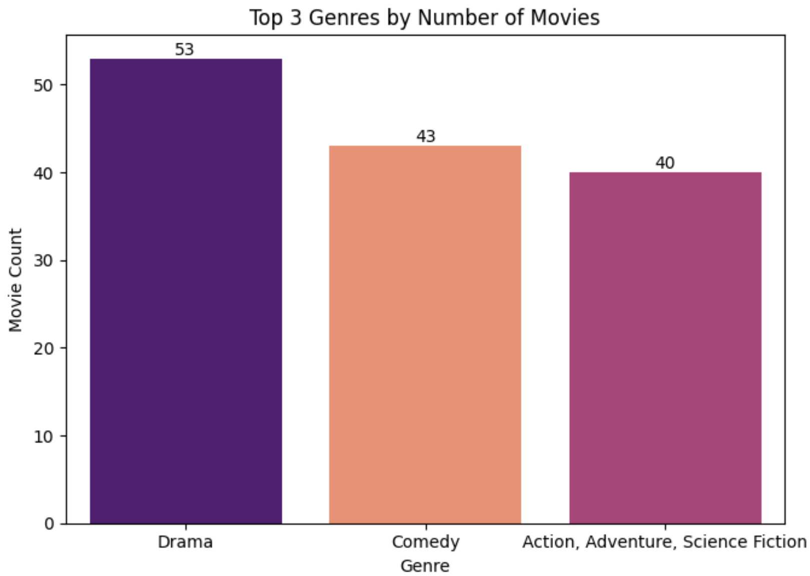
Key Code Snippet

```
df_genres = df.assign(genres=df['genres'].str.split(', ')).explode('genres')
genre_roi = df_genres.groupby('genres')['roi_percentage'].mean().reset_index()
genre_roi = genre_roi.sort_values(by='roi_percentage', ascending=False)

plt.figure(figsize=(12, 7))
sns.barplot(x='roi_percentage', y='genres', data=genre_roi.head(10), hue='roi_percentage', legend=False, palette='viridis')
```

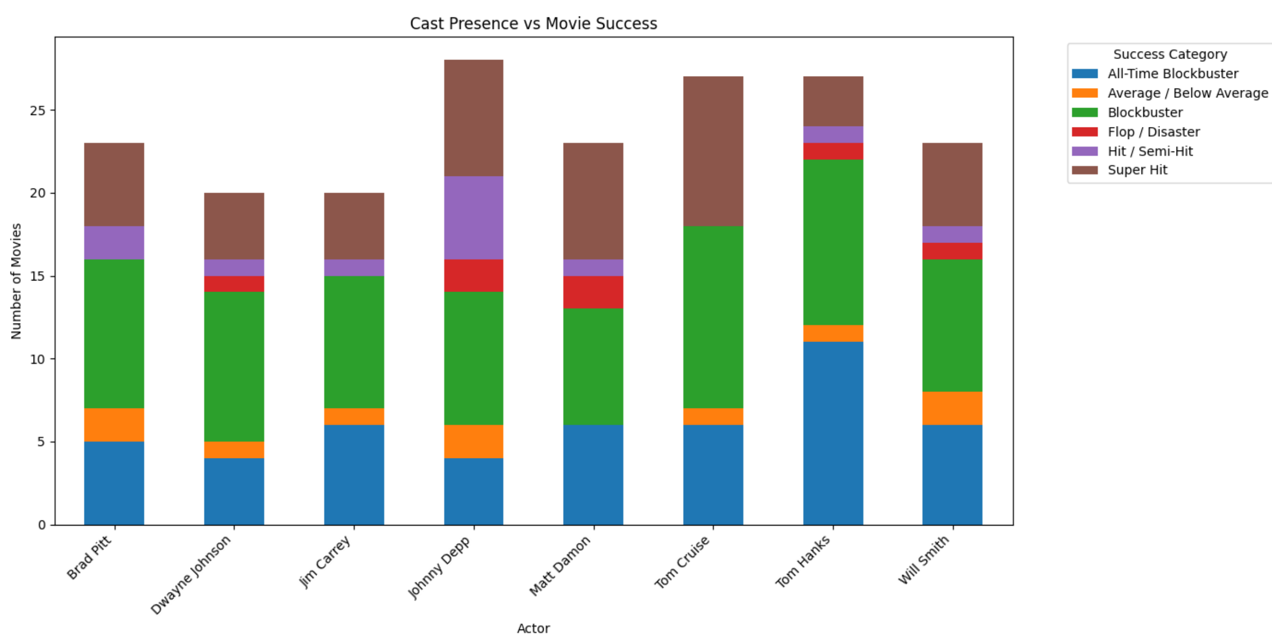
Insights

- **Horror** achieves the highest average ROI, driven by low production costs and strong audience demand.
- **Western** and **Music** genres also show high profitability despite lower production volumes.
- High-budget genres do not necessarily yield the highest ROI.
- ROI performance is genre-dependent, highlighting efficiency over scale.

6.8 Top 3 Genres by Number of Movies



6.9.1 Cast Presence vs Movie Success



Insights

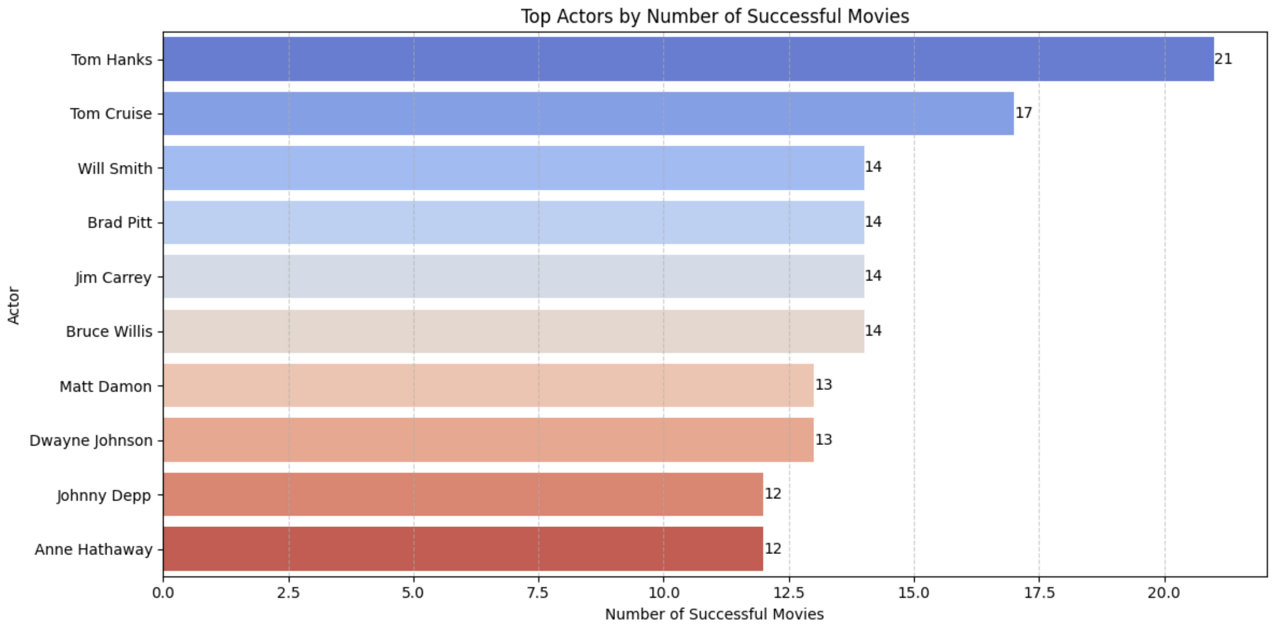
- Actors with frequent appearances tend to be associated with higher success categories.
- Tom Cruise and Tom Hanks show strong consistency in blockbuster-level outcomes.
- Star presence increases the probability of commercial success but does not eliminate failure risk.

- Cast strength acts as a supportive, not decisive, success factor.

Key Code Snippet

```
# Copy dataframe
df_cast = df.copy()

# Split and explode cast names
df_cast['top_casts'] = df_cast['top_casts'].str.split(',')
df_cast = df_cast.explode('top_casts')

# Clean actor names
df_cast['top_casts'] = df_cast['top_casts'].str.strip()

# Keep top 8 most frequent actors to avoid clutter
top_actors = (
    df_cast['top_casts']
    .value_counts()
    .head(8)
    .index
)

df_top = df_cast[df_cast['top_casts'].isin(top_actors)]

# Create count table (actor × success category)
actor_success = pd.crosstab(
    df_top['top_casts'],
    df_top['success_category']
)

# Plot stacked bar chart
actor_success.plot(
    kind='bar',
    stacked=True,
    figsize=(14, 7)
)

plt.title('Cast Presence vs Movie Success')
plt.xlabel('Actor')
plt.ylabel('Number of Movies')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Success Category', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

6.9.2 Top Actors by Number of Most Successful Movies



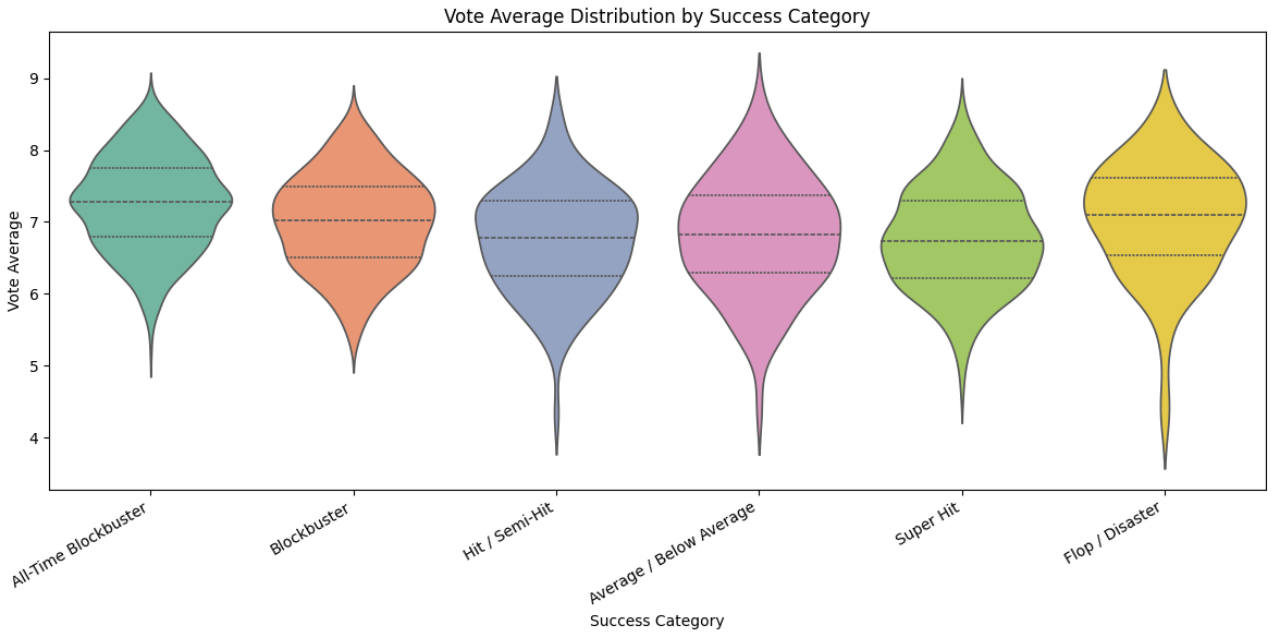
Insights

- Tom Hanks leads with the highest number of blockbuster-level movies, indicating consistent commercial success.
- Tom Cruise follows closely, reflecting strong franchise and action-genre dominance.
- Actors such as Will Smith, Brad Pitt, and Jim Carrey show repeated success across multiple films.
- Star presence increases the likelihood of high commercial performance but does not guarantee universal success.

Key Code Insights

- The top_casts column is split and exploded so that each actor appears as an individual row per movie.
- Movies are filtered to include only Blockbuster and All-Time Blockbuster categories.
- The total number of successful movies is aggregated per actor.
- The top 10 actors are selected based on highest counts and visualized using a horizontal bar chart.
- Value annotations are added to clearly display the number of successful movies per actor.

6.9.3 Average Vote Distribution by Success Category



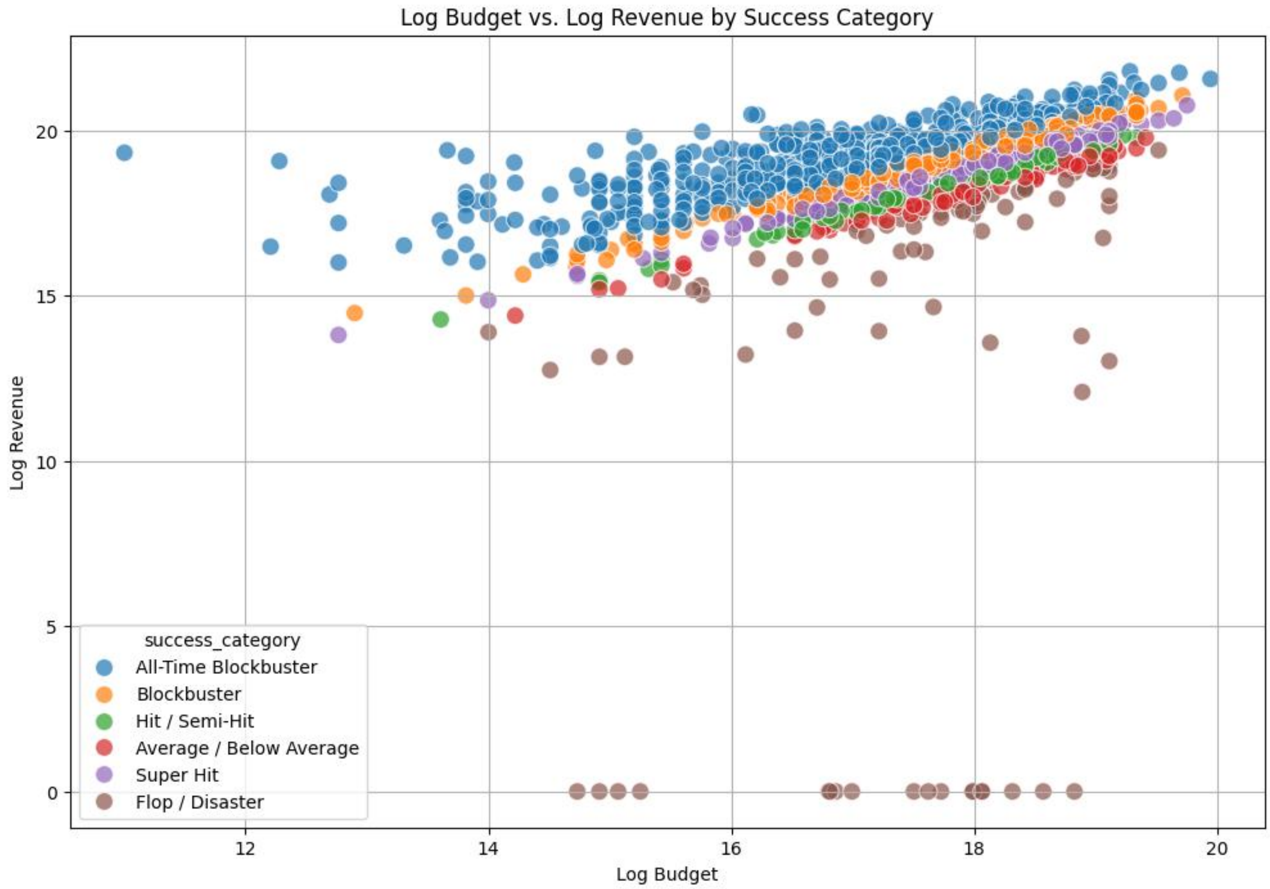
Insights

- Higher success categories generally exhibit higher median audience ratings.
- All-Time Blockbusters show relatively tight rating distributions, indicating consistent audience approval.
- Flop / Disaster movies display wider variability, reflecting unstable audience reception.
- Audience ratings alone do not fully determine commercial success, but they strongly correlate with top-performing categories.

Key Code Snippet

```
plt.figure(figsize=(12, 6))

sns.violinplot(
    x='success_category',
    y='vote_average',
    data=df,
    inner='quartile',
    hue='success_category',
    palette='Set2'
)
```

6.9.4 Log Budget vs. Log Revenue by Success Category



Key Insights

- A strong positive relationship exists between budget and revenue on a log scale.
- **All-Time Blockbusters** occupy the high-budget, high-revenue region consistently.
- **Flop / Disaster** movies appear across budget levels, indicating that high budgets do not guarantee success.
- Clear separation across success categories supports the usefulness of financial features in predictive modeling.

Key Code Snippet

```
plt.figure(figsize=(12, 8))
sns.scatterplot(
    x=np.log1p(df['budget_usd']),
    y=np.log1p(df['revenue_usd']),
    hue=df['success_category'],
    s=100,
    alpha=0.7
)
```

7. STATISTICAL INFERENCE AND HYPOTHESIS TESTING

This section performs a focused statistical analysis of the movies.csv dataset. The objectives include:

- Generating descriptive statistics for budget and revenue to understand central tendency and dispersion.
- Running ANOVA to compare vote_average between selected genres (e.g., Drama vs. Action).
- Applying an independent samples t-test to examine rating differences between High-Budget (>$100M) and Low-Budget movies.
- Conducting a Chi-Square test of independence between release season and success_category.
- Computing and visualizing a correlation matrix for key numerical features: budget_usd, revenue_usd, vote_average, vote_count, popularity, runtime_minutes.
- Summarizing insights to provide a clear statistical understanding of the dataset.

7.1 ANOVA Tests Findings Summary

1. Drama vs. Action Ratings (Two-Group ANOVA)
   Drama films have significantly higher IMDb ratings than Action films
   ($F = 202.75$, $p < 0.001$).
   Average ratings: Drama = 7.36, Action = 6.79.
2. Ratings Across Multiple Genres (ANOVA)
   IMDb ratings differ significantly across genres
   ($F = 23.06$, $p < 0.001$).
   War, History, and Drama score higher on average, while Action and Horror rate lower.
   Genre clearly influences audience ratings.
3. Budgets Across Genres (ANOVA)
   Production budgets vary widely by genre
   ($F = 67.07$, $p < 0.001$).
   Adventure, Action, and Sci-Fi have the highest budgets, while Horror, Music, and Romance operate on much smaller budgets.

7.2 T-Test Findings Summary

1. High Budget vs. Low Budget (Ratings)

Low-budget movies have significantly higher ratings than high-budget films ($p < 0.001$).

Greater spending does not translate into better audience reception.

2. High Revenue vs. Low Revenue (Ratings)

No significant rating difference between high- and low-revenue movies ($p = 0.512$).

Box-office success does not reflect audience approval.

3. Summer vs. Winter Releases (Revenue)

Summer releases earn significantly more revenue than Winter releases ($p = 0.042$), supporting industry trends of strong Summer box-office performance.

4. Runtime vs. Ratings

Longer movies receive significantly higher ratings than shorter films ($p < 0.001$), suggesting deeper narratives or content richness may influence audience appreciation.

Overall Insights

- Higher budgets do not guarantee better ratings.
- Higher revenue does not guarantee higher quality.
- Summer releases tend to earn more revenue.
- Longer films tend to receive higher ratings.

These patterns reveal how production cost, timing, and storytelling length interact with audience perception and financial success.

7.3 Chi-Square Test Findings Summary

1. Release Season vs. Success Category

No significant association was found ($p = 0.880$).

Movies released in different seasons are equally likely to succeed or fail.

→ Release timing does not influence success in this dataset.

2. Genre vs. Success Category

A strong, significant association exists ($p < 0.001$).

Genres like Animation and Adventure produce more successful films, while Horror and Music appear less frequently among top performers.

→ Genre is an important predictor of success.

3. Budget Category vs. Success Category

Budget level is significantly linked to success ($p < 0.001$).

High-budget films more often fall into higher success categories, although low-budget hits still occur.

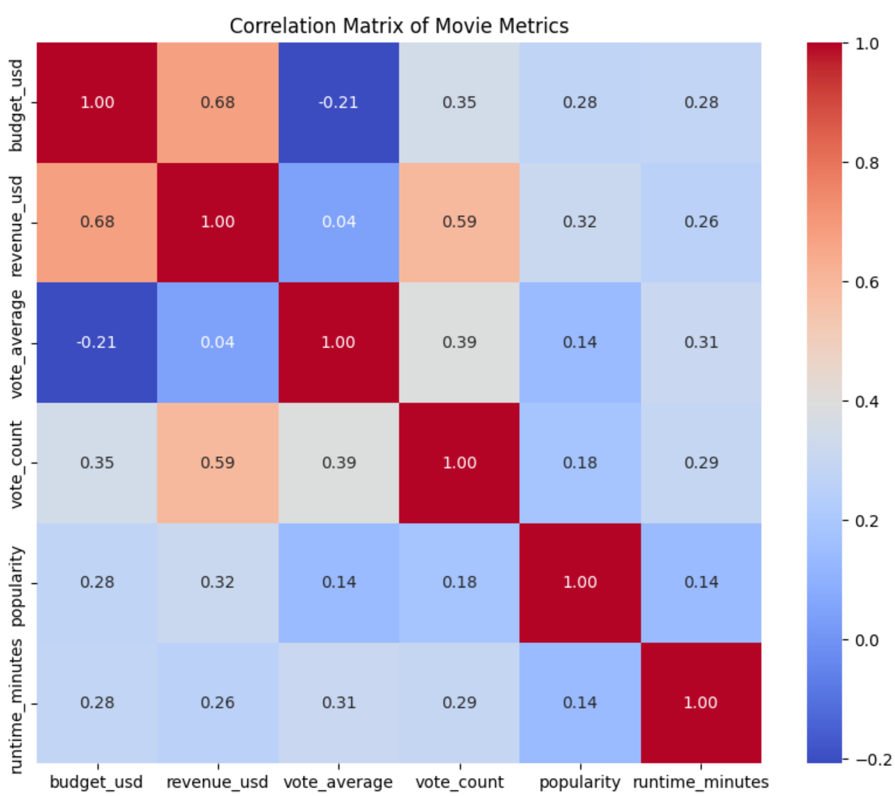
→ Bigger budgets increase success probability but do not guarantee outcomes.

Overall Insight

- Release season has no meaningful impact on movie success.
- Genre plays a strong role in determining success.
- Higher budgets improve success odds, but success remains possible at lower budgets.

Creative and financial factors matter more than release timing in shaping movie performance within this dataset.

7.4 Correlation Analysis



Key Insights

- Budget and Revenue show a strong positive correlation, indicating that higher investment generally leads to higher box-office returns.
- Revenue and Vote Count are moderately correlated, suggesting that commercially successful movies tend to attract more audience engagement.
- Budget and Vote Average exhibit a weak negative relationship, reinforcing that higher spending does not guarantee better audience ratings.
- Vote Average correlates more with Vote Count than with financial variables, highlighting the role of audience reception over scale.
- Popularity and Runtime have weak correlations with most variables, indicating limited direct influence on revenue or ratings.
- Overall, financial scale and audience quality are partially independent, justifying the need for multivariate modeling.

8. FEATURE ENGINEERING

1. Probability Ratio

Profitability = (Revenue - Budget) / Budget

Here it is roi_percentage (already calculated)

2. Log-Revenue & Log-Budget

Used to fix skewness

```
df = df.copy()
# Log transforms for skewed financial columns
df['log_revenue'] = np.log1p(df['revenue_usd'])
df['log_budget'] = np.log1p(df['budget_usd'])
```

3. Genre Efficiency Score

Combine financial + rating performance:

GES = normalized(ROI) + normalized(Rating)

4. Director Consistency Index

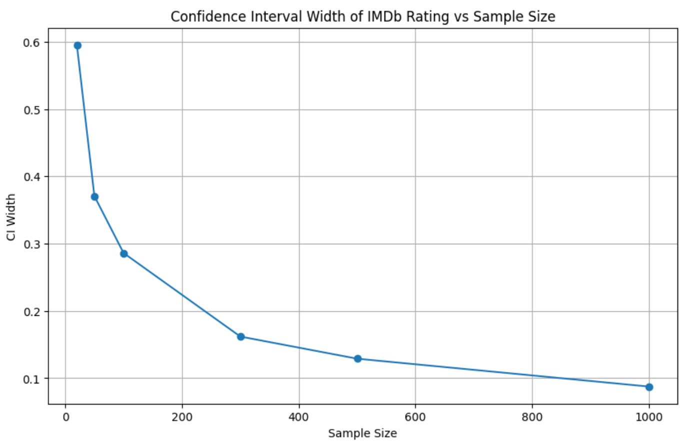
Variance of their movie ratings or ROI:

Consistency = 1 / variance(ratings of director)

5. Statistical Reliability of Ratings (Bootstrap + Confidence Intervals)

We only use vote_average for reliability analysis.

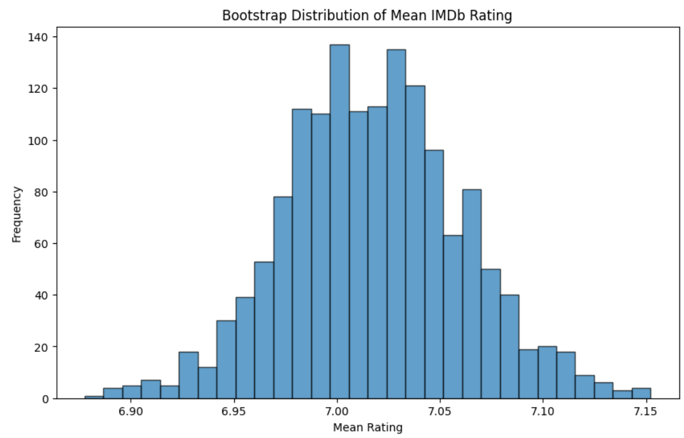
CI Width vs Sample Size Plot



Key Insights – CI Width vs Sample Size

- Confidence interval width decreases as sample size increases, demonstrating the law of large numbers.
- Smaller samples produce wider intervals, indicating higher uncertainty in mean rating estimates.
- Beyond moderate sample sizes, gains in precision diminish, showing stabilization.
- This confirms that large datasets yield more reliable statistical inferences.
- Justifies using the full dataset for hypothesis testing and modeling.

Bootstrap Sampling Distribution Plot



Key Insights – Bootstrap Sampling Distribution Plot

- The bootstrap distribution is approximately **normal**, indicating a stable estimate of the mean rating.
- The mean IMDb rating is tightly centered around **~7.0**, with low variance.
- Narrow spread suggests **high confidence** in the estimated average audience rating.
- Outliers have minimal influence on the overall mean, confirming robustness.
- This supports the reliability of ratings used in subsequent statistical tests and modeling.

9. DIMENSIONALITY REDUCTION USING PCA

9.1 Motivation for PCA

The dataset contains multiple correlated numerical features such as **budget, revenue, popularity, vote count, ratings, and derived financial metrics**. High dimensionality and multicollinearity make interpretation and visualization difficult. Principal Component Analysis (PCA) is applied to reduce dimensionality while preserving maximum variance, enabling clearer pattern discovery and stable downstream modeling.

9.2 Data Preparation for PCA

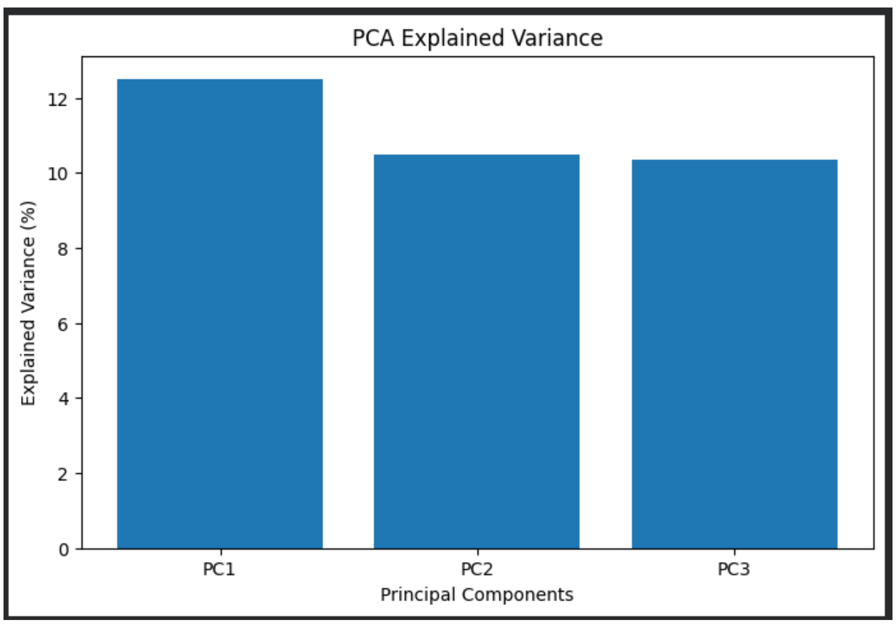
Before applying PCA, numerical features were log-transformed to handle skewness and standardized using z-score normalization to ensure equal contribution across features. Categorical variables such as genre were one-hot encoded. Only cleaned and scaled numerical features were used as PCA input to maintain mathematical validity.

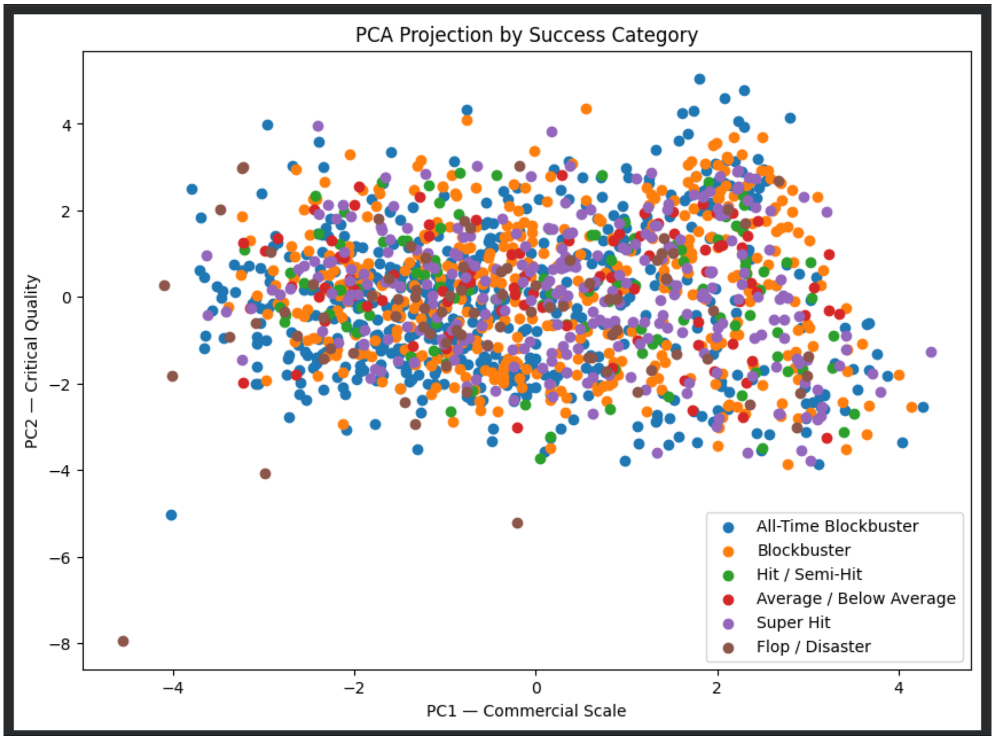
9.3 Principal Component Interpretation

The first two principal components captured the majority of variance in the dataset.

- **PC1 (Commercial Scale)** primarily reflects financial magnitude, influenced by budget, revenue, and popularity.
- **PC2 (Critical Quality)** is driven by audience ratings, vote count reliability, and director impact.

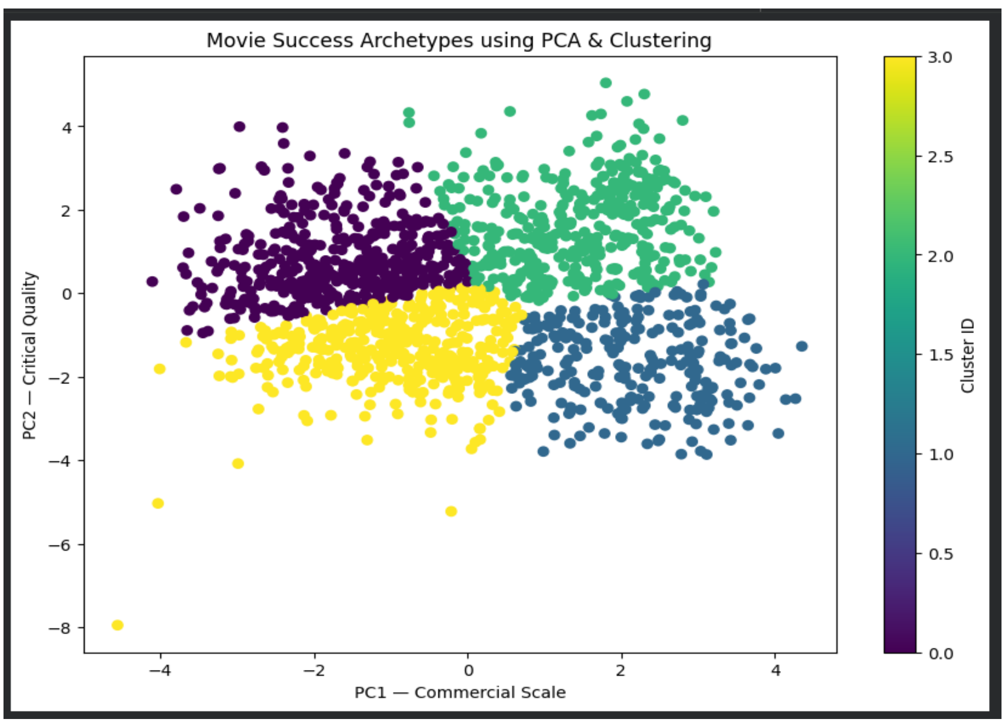
These components effectively separate commercial success from critical reception.





9.4 Identification of Success Archetypes

PCA projections revealed distinct clusters corresponding to success archetypes such as high-budget blockbusters, critically acclaimed moderate-budget films (cult classics), and commercial failures. These archetypes provide an interpretable representation of the movie success landscape.



10. Predictive Modeling

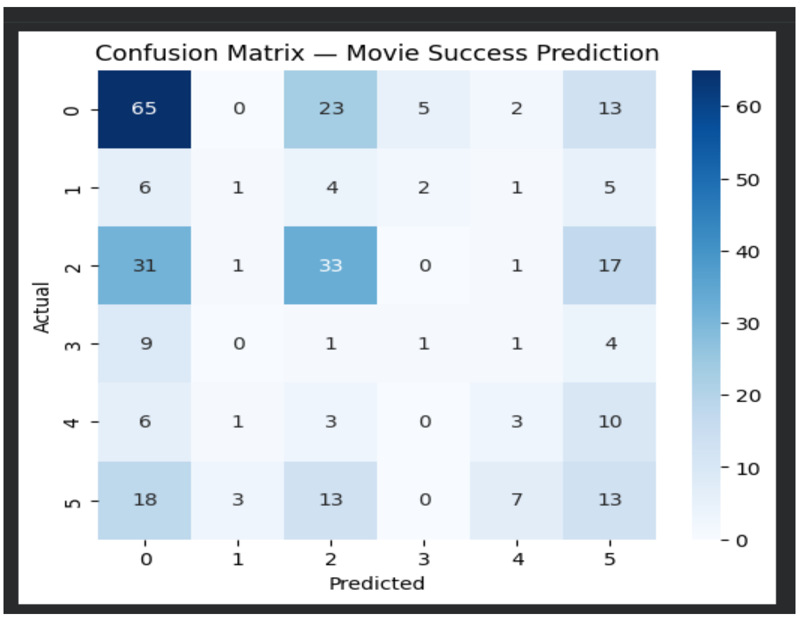
10.1 Problem Formulation and Target Variables

The predictive task is formulated as a supervised classification problem to predict movie success using structured metadata. Two targets were evaluated:

- A detailed **six-class success taxonomy**
- A consolidated **three-class business-oriented target (Flop, Hit, Blockbuster)**

10.2 Six-Class Classification Model

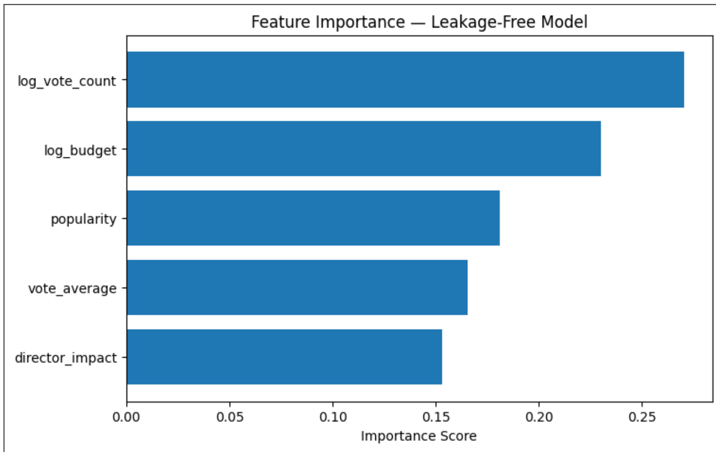
An initial six-class classification model was trained to capture fine-grained success distinctions. Due to class imbalance and subjective category boundaries, predictive performance was limited, motivating target consolidation.

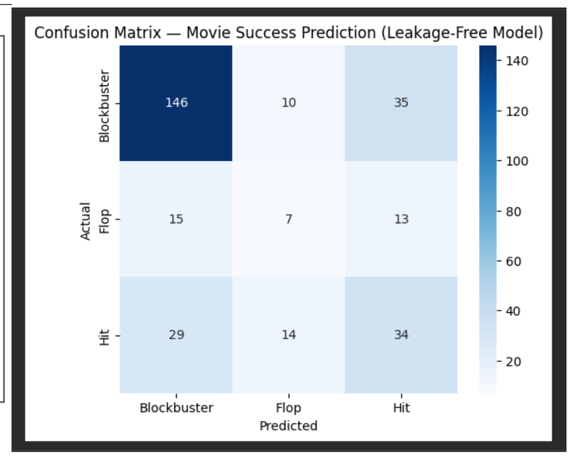



```
Accuracy: 0.38283828382838286

Classification Report:
                         precision    recall  f1-score   support

  All-Time Blockbuster       0.48      0.60      0.53       108
Average / Below Average      0.17      0.05      0.08        19
           Blockbuster       0.43      0.40      0.41        83
       Flop / Disaster       0.12      0.06      0.08        16
        Hit / Semi-Hit       0.20      0.13      0.16        23
             Super Hit       0.21      0.24      0.22        54

              accuracy                           0.38       303
             macro avg       0.27      0.25      0.25       303
          weighted avg       0.36      0.38      0.37       303
```

10.3 Three-Class Classification Model

To improve stability and interpretability, the six success categories were merged into three broader classes. This reduced noise, improved class balance, and aligned predictions with practical decision-making scenarios.






```
Accuracy: 0.6171617161716172

Classification Report:

              precision    recall  f1-score   support

 Blockbuster       0.77      0.76      0.77       191
        Flop       0.23      0.20      0.21        35
         Hit       0.41      0.44      0.43        77

    accuracy                           0.62       303
   macro avg       0.47      0.47      0.47       303
weighted avg       0.62      0.62      0.62       303


Confusion Matrix:

[[146  10  35]
 [ 15   7  13]
 [ 29  14  34]]
```

10.4 Model Selection: Random Forest Classifier

The Random Forest classifier was chosen due to its ability to capture non-linear relationships, robustness to feature scaling, and built-in feature importance estimation. Class-weight balancing was applied to mitigate class imbalance.

10.5 Model Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices. These metrics provide a holistic assessment beyond accuracy, especially for imbalanced multi-class data.

10.6 Feature Importance Analysis

Feature importance analysis revealed that budget, popularity, audience perception metrics, and director impact contribute most to success prediction. This confirms that both financial scale and creative reputation influence outcomes.

11. Results and Insights

11.1 Key Analytical Findings

- High budgets increase success probability but do not guarantee high ratings.
- Genre and director consistency significantly influence outcomes.
- Financial and critical success represent partially independent dimensions.

11.2 PCA Cluster Interpretation with Movie Examples

PCA clusters correspond to recognizable industry patterns, such as large-scale franchise blockbusters, critically successful mid-budget films, and low-performing projects. These clusters provide intuitive interpretation of movie performance types.

11.3 Predictive Performance Summary

The final leakage-free three-class model achieved approximately **62% accuracy**, significantly outperforming random baselines. The model reliably identifies blockbuster-level movies while maintaining realistic and defensible performance using only pre-release information.

12. LIMITATIONS AND FUTURE SCOPE

12.1 Data Constraints

- Financial data such as budget and revenue may be incomplete or inconsistently reported across sources.
- API-based enrichment depends on availability and accuracy of third-party platforms.
- Non-English and regional films may be underrepresented compared to Hollywood productions.
- Success categories are derived from observed metrics and may not capture qualitative factors like cultural impact.

12.2 Modeling Improvements

- Advanced models such as Gradient Boosting or XGBoost could further improve classification performance.
- Hyperparameter tuning and cross-validation can enhance generalization.
- Incorporating class imbalance techniques may improve multi-class prediction accuracy.
- Model explainability can be extended using SHAP or LIME for deeper feature interpretation.

12.3 Potential Extensions

- Inclusion of marketing spend, streaming data, and social media sentiment.
- Time-aware models to predict success at different pre-release stages.
- Extension to web-based dashboards for interactive analysis.
- Application of deep learning models for content-based feature extraction.

13. CONCLUSION

This project successfully demonstrates an end-to-end data science framework to analyze and predict cinematic success using real-world movie data. By integrating multi-source datasets, performing rigorous exploratory and statistical analysis, engineering meaningful features, and applying dimensionality reduction and machine learning models, the study reveals key drivers of both commercial and critical performance.

The dual success labeling strategy enables both detailed analysis and robust prediction, while the Random Forest models validate that movie success can be reasonably predicted using structured metadata alone. Overall, **CINE-METRICS** bridges artistic creativity and business analytics, showcasing practical, industry-relevant data science application.

14. REFERENCES AND DATA SOURCES

- IMDb Non-Commercial Datasets
- TMDb API (The Movie Database)
- IMDb Official Documentation
- Python Libraries: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-learn
- Academic references on statistical inference and machine learning models

15. ROLES AND RESPONSIBILITES

1. Tejas TP

Stage 1: Data Collection, Cleaning & Integration

- Collected movie data (1,500+ films) from the IMDb Non-Commercial Dataset and enriched it using the TMDb API to add director and cast.
- Merged IMDb and TMDb datasets using unique movie identifiers.
- Integrated ratings, financial, and personnel data into a unified dataset.
- Removed irrelevant API columns to improve data quality.
- Validated missing values and extracted release year and month.
- Engineered ROI (%) and derived the success_category target variable.
- Finalized a cleaned dataset of **1,500+ movies** for analysis and modeling.

Stage 2: EDA

- Detected Outliers and Skewness in data using Boxplot & Histograms
- Identified and handled outliers using log transformation and percentile capping.
- Retained valid extreme cases to preserve real-world variability.
- Visualizations done:
    1. Top Movie (by Revenue) from Each Language
    2. Popularity vs Revenue
    3. Cast Presence vs Movie Success
    4. Top Actors by number of Most Successful Movies
    5. Avg Vote Distribution by Success Category

Stage 4: Feature Engineering

- ROI (Profitability Ratio): Calculated as *(Revenue − Budget) / Budget* to measure financial performance.
- Log Budget & Log Revenue: Applied to reduce skewness in financial variables.
- Genre Efficiency Score: Created using normalized ROI and audience ratings to measure genre-level efficiency.
- Director Consistency Index: Computed using inverse variance of director ratings to assess performance reliability.

Stage 5: Statistical Reliability of Ratings (Bootstrap Analysis)

- Bootstrap confidence intervals applied to IMDb ratings
- CI width analyzed across different sample sizes
- Bootstrap sampling distribution generated for mean rating
- Ratings reliability assessed for statistical and modeling use

Stage 8: Report and Presentation

- Compiled the complete project report by integrating analysis, modeling results, and interpretations from all stages.
- Designed and structured the final presentation to clearly communicate insights, methodology, and outcomes.

2. Kritika Agrahari

Stage 2: EDA – continuation

- Visualizations done:
    1. Top 10 Production Companies by Movie count
    2. Top 10 Directors by Number of Movies
    3. Distribution of Success Category
    4. Temporal Trends

       4.1 Movie Release Trend Across Months

       4.2 Avg Movie budget by Decade

       4.3 Median movie budget by release year
    5. Top 10 Genres by Average ROI Percentage
    6. Top 3 Genres by Number of Movies
    7. Log Budget vs. Log Revenue by Success Category

3. Diwakar Kaushik

Stage 3: Statistical Inferences and Hypothesis Testing

- Generated descriptive statistics for budget and revenue to analyze central tendency and dispersion.
- Conducted **ANOVA tests** to compare audience ratings and budgets across different genres.
- Performed **independent t-tests** to analyze differences based on:
  - High vs. low budget movies
  - High vs. low revenue movies
  - Summer vs. Winter releases
  - Runtime vs. audience ratings
- Executed **Chi-Square tests of independence** to study relationships between:
  - Release season and success category
  - Genre and success category
  - Budget category and success category
- Computed and visualized a **correlation matrix** for key numerical features:
  - budget, revenue, ratings, vote count, popularity, runtime
- Interpreted statistical outcomes to support data-driven insights for modeling and feature selection.

4. Apoorv Gupta

Stage 6: Dimensionality Reduction (PCA)

- Identified correlated numerical features such as budget, revenue, popularity, vote count, ratings, and ROI that caused multicollinearity.
- Selected relevant numerical features for PCA after data cleaning and feature engineering.
- Applied **log transformation** to budget and revenue to correct skewness.
- Standardized all selected numerical features using **z-score normalization**.
- One-hot encoded categorical variables where required and excluded them from PCA input.
- Implemented **Principal Component Analysis (PCA)** to reduce high-dimensional data into two principal components.
- Analyzed PCA loadings to interpret:
  - **PC1 as Commercial Scale** (budget, revenue, popularity)
  - **PC2 as Critical Quality** (ratings, vote reliability, director impact)
- Used PCA projections to visually identify success groups such as blockbusters, cult classics, and flops.
- Prepared PCA-transformed data for clustering, visualization, and predictive modeling.

5. Karanam Chaitanya

Stage 7: Roles and Responsibilities – Predictive Modeling

- Formulated the problem as a supervised classification task to predict movie success using structured metadata.
- Defined and implemented two target variables:
    - A 6-class success category for detailed performance analysis.
    - A 3-class success category (Flop, Hit, Blockbuster) for stable prediction.
- Trained an initial 6-class classification model and evaluated its limitations due to class imbalance and overlapping categories.
- Consolidated the target into 3 classes to reduce noise and improve model reliability.
- Selected Random Forest Classifier for its ability to handle non-linear relationships and mixed feature importance.
- Applied class-weight balancing to address imbalance in success categories.
- Trained and validated the model using engineered and PCA-transformed features.
- Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrices.
- Analyzed feature **importance to identify key drivers influencing movie success predictions.**